Stablecoin On / Off-Ramp — System Architecture

save in stablecoins · spend on local rails · cash out in minutes — always know fee, ETA & risk before you confirm

USER
save · spend · cash out

opens app

LAYER A · CONSUMER APP

the only surface that exposes certainty — never complexity

- Onboarding & KYC tier
- Wallet balance
- Save / Spend / Cash-out
- Quote screen
- Payment instructions
- Live order tracking
- History
- Support & Dispute
- Trust labels
- Limits & tier
- React Native — mobile app
- Next.js — web dashboard / landing

one quote · one CTA · one timeline · one support path

request quote / confirm

API GATEWAY · AUTH · SESSION · RATE-LIMIT — every request passes through here

screen

authenticated request

CORE SERVICES · the operating brain

internal complexity → external simplicity

score

- **Quote Engine** — price + fee + rate · ETA · 45s lock
- **Routing & Matching** — picks best merchant by fill · speed · spread · trust · auto-failover
- **Order State Machine** — orchestrates every order + every state
- **Wallet Service** — embedded / connected balances · transfers
- **Notification Svc** — live status push · reminders · webhooks

match · lock escrow

LAYER B · MERCHANT / LIQUIDITY

who actually fills orders — recruited, scored, bonded

- KYB & due diligence
- Float & inventory
- Payout rails: UPI / Pix / bank
- Response SLAs
- Performance scoring
- Dispute handling
- Reserve / bond
- Spread quoting

fulfil → release

LAYER E · SETTLEMENT & PROOF (on-chain · Base)

monitor

proves what happened — used as infrastructure, never as interface

- Escrow & conditional release
- Merchant registry & staking
- Proofs / attestations
- Audit log
- Pause / circuit breaker
- RBAC · multisig admin
- On-chain events

persist · observe

INFRASTRUCTURE & OBSERVABILITY

DATA
- PostgreSQL — transactional
- Redis — queues · locks · quote cache
- Event-driven order bus

OBSERVE
- Full audit trail
- Latency monitoring
- Failure-state alerts
- Reconciliation dashboards
- Fraud analytics

LAYER D · COMPLIANCE & RISK

the guard — mostly invisible, always on, wraps every layer

- Progressive risk tiers (KYC)
- Wallet / address screening
- Transaction monitoring
- Fraud scoring
- Velocity checks
- Geo restrictions
- Pattern detection
- Merchant behaviour scoring

COLOR KEY

- Consumer · what users see
- Core services · the brain
- Routing · the moat
- Merchant · liquidity
- On-chain · settlement
- Compliance & risk · guard
- Data & observability

Design rule: build a MONEY PRODUCT with protocol advantages — not a protocol product with money UX.

every extra user decision is a UX tax · expose certainty, hide complexity · failure-first, then growth

Transaction Lifecycle — the order state machine

happy path across the top · failure-first branches below — design every FAILED flow before the growth flow

HAPPY PATH

one quote · one CTA · one timeline — the user always sees exactly where they are

1. **Quote locked** — rate · fee · ETA valid 45 s
2. **Pay instructions** — exact amount + reference code
3. **Waiting for payment** — user pays via local rail
4. **Payment detected** — amount + ref auto-matched
5. **Merchant confirms** — fiat payout / receipt verified
6. **Funds released** — USDC released · payout sent
7. ✓ **Completed** — repeatable in 1 tap next time

any state can branch on failure ↓

WHEN THINGS GO WRONG · failure-first

the best dispute system is the one users rarely need

THINGS THAT CAN FAIL

- quote expires mid-flow
- user paid wrong amount
- paid from unsupported account
- payment rail is down
- stablecoin transfer delayed
- merchant times out
- merchant claims no receipt
- bank freezes merchant account
- user claims no payout
- risk threshold crossed mid-txn

FOR EACH FAILURE, DEFINE

1. Detection method
2. User-facing message
3. Automated next step
4. Manual escalation path
5. Liability owner
6. Resolution SLA

resolve

SYSTEM RESPONSE / RESOLUTION

every failure resolves to one of four visible states

- **Delayed** — ETA updated · user told · reminder sent → resumes happy path
- **Needs action** — wrong amount / account · one clear instruction → user fixes · resumes
- **Under review** — proofs + attestations · manual escalation → resolved or refunded
- **Refunded / reversed** — liability owner makes the user whole → terminal · made whole

DISPUTE PREVENTION > dispute resolution

PREVENT

- exact amount matching
- reference-code checks
- name / account confirm
- order expiry timers
- merchant response deadlines
- duplicate-payment detection
- anomaly alerts
- auto reminders
- instant escalation when a state stalls

A user must never wonder: "did my money disappear?" — every state answers: what's happening · what to do · who's responsible · ETA

Trust, Compliance & Routing

the invisible machinery

heavy checks kill conversion · skipping them breaks the system — so trust is progressive and routing is automatic

PROGRESSIVE KYC · tiered trust

↑ verify more to unlock more — friction scales with risk, not with everyone

0 · Tier 0 · Browse & simulate

view-only

see live quotes, simulate flows — no money moves

- no KYC
- just open the app

1 · Tier 1 · Get started

low daily cap

first real transactions at low limits

- basic wallet
- device checks
- minimal identity signal

2 · Tier 2 · Higher limits

↑ limits

bigger orders, more local rails unlocked

- govt ID
- liveness
- local payment verification

3 · Tier 3 · High-value

high-value

large tickets, source-of-funds backed

- source of funds
- enhanced screening
- manual review if needed

M · Merchants · full KYB

bonded LP

join the liquidity network & fill orders

- KYB
- bank verification
- treasury source
- reserve / bond
- ongoing monitoring

MATCHING ENGINE · the moat

a marketplace, not a DEX — rank every merchant, route automatically, fail over instantly

RANKING SIGNALS

- success rate
- online now
- available float
- payout rail
- spread
- avg completion
- queue load
- dispute rate
- fraud flags
- bank health
- geo fit

every merchant scored in real time

ROUTING ENGINE

fill · speed
spread · fraud
auto-failover

USER JUST SEES

- Fastest
- Lowest fee
- Highest reliability
- Best for large orders
- Best for first-timers

complex scoring underneath, plain-English labels on top

default UX: "best route selected for you" — advanced users can still pick manually

THE QUOTE SCREEN · highest-leverage UI

‹ Quote — valid 45s

You send
INR 5,000

You receive
58.42 USDC

Rate	1 USDC = 85.6 INR
Fee	₹40
Network fee	₹8
Total time	2 – 4 min
Payout method	UPI
Route quality	High confidence

Continue | Edit

WHY THIS SCREEN CONVERTS

- → send + receive both shown — no mental math
- → every fee itemized — nothing hidden
- → ETA range sets honest expectations
- → route quality = plain trust signal
- → 45s lock — real price, never stale
- → two choices only: continue or edit

certainty before you confirm

no clutter · no jargon · no surprises

Tiered trust keeps onboarding light · automatic routing hides the marketplace · the quote screen earns the tap — complexity in, simplicity out